Stability Analysis of Car Driving with a Joystick Interface

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***Abstract*—This paper presents stability analysis of car driving with a joystick interface. A car drive assist system with a joystick interface enables disabled persons to drive a car with joystick operations. A steering wheel and gas/brake pedals are actuated by electric motors which rotations are controlled by micro-computers based on 2DOF joystick commands. Since the drive assist system involves a time delay and an amplifier with a high gain, a car would show unstable behaviors when a stable condition is not guaranteed. To analyze the stable and unstable behaviors of car driving, we derive a vehicle dynamic model together with the joystick system and human recognition model (driver model) for various car velocities.**

Keywords—stability analysis; car driving; joystick; disabled persons;

I. Introduction

Standard cars are designed for persons who can operate normal control devices such as gas and brake pedals, a steering wheel and a shift lever by their hands and legs. However it is quite difficult for disabled persons with handicaps on their legs or arms to drive standard cars without any modifications or special devices.

Since those cars are manufactured by mass-production processes, to change its basic construction for customizing purpose is inefficient and its cost would be extremely increased.

To overcome this problem, a joystick system was researched and tested in Japan but it was implemented on EV and any successful results could not be obtained[1]. The commercial available hydraulic system based joystick system [2] is more widely applicable however, it has limitation in applicable car types since modern cars equip with electrical power steering systems rather hydraulic ones. The electric motor based joystick system [3] is commercial available however, the joystick interface is too small and too sensitive for vehicle high speed driving.

To provide a joystick car drive system with enhanced safety and driving comfort, we are now developing an add-on type joystick car drive system with advanced function for safety and comfort [4].

In this paper, we discuss about the condition for driving a car with a joystick interface to maintain stability in the high speed drive. To realize the high speed drive, we propose a variable gain method in a steering control system. A velocity depending variable gain between a joystick angle and that of a steering wheel enables a stable car driving in a wide range of speed with no stress on a driver. The availability of the proposed method is validated by computer simulations.

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II. Car Drive Assist System with Joystick Interface

A. System

Fig.1 shows an overview of a car cockpit with a joystick drive system. The joystick car drive system consists of two subsystems. One is for steering-wheel actuating and the other is for pedal actuating. Each subsystem has almost an identical mechanism and a control system. Each system equips an individual microprocessor to realize control functions specified for each operation. Two subsystems allow a handicapped person to drive a car by using two joysticks. The right joystick can be operated in the lateral direction to control a steering wheel rotation, while the left joystick can be operated in longitudinal direction to activate a gas pedal and a brake pedal.

The joystick interface can be operated in a small force with a short stroke. However when a vehicle running at a high speed, a wrong joystick operation, such as a sudden and wide steering, results in a car accident quite easily. A quite sensitive steering control is required around the center of steering angle when a car runs along a straight line. The other hand, when parking a car, a large steering angle provides easier parking maneuvering to a driver. Additionally, the joystick system has a time delay because of the mechanical inertia and friction and computer calculation time. This time delay influence to a drive stability, especially running at a high speed.

To satisfy these multiple requirements and solve some problems for the stability on the joystick operated vehicle system, we propose a steering control method with variable steering sensitivity function based on vehicle velocities. To realize these functions, a dynamic model of a car with joystick system is analyzed.

To realize stable driving in a wide range of vehicle speed, variable steering sensitivity function is proposed which varies a gain between the joystick inclination angle to a rotation angle of a steering wheel.

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B. *A joystick system as a human-machine interaction*

By the joystick system, a driver is completely separated physically from drive operation devices, such as a steering-wheel and pedals. This type of drive system is so called "by-wire" in the automobile field, however it is regarded as a teleoperation system in the robotic field which has synergistic relationships with CogInfoCom. A computer system is inserted between a human and a machine which can transfer information from a human side to a machine and/or from a machine side to a human. In car driving applications since human moves with a car, therefore he/she can get a lot of other information from a car, such as speed, acceleration, rotation, etc. Also visual information of the environment can be obtained directly.

There are some types of teleoperation system (such as [5]). An unilateral system enables a simple control architecture while bilateral systems with/without a force feedback brings advanced operation feeling to a human operator.

In this paper, we study basis of a joystick system for a car driving application. Therefore we assume that the unilateral control is applied to a joystick system in which a position command is sent to a slave mechanism (steering- or pedal-drive mechanism in this case) with position feedback whose reference can be scaled by the computer. The scaling affects a sensitivity of a steering operation which results in a driving stability of a car at variety of the speed range. We will investigate the relationships in this paper.

Fig.1 An installation example of a two-lever type joystick system

III. MODELING

A. *Model of a car with joystick system*

Figure 2 shows a dynamic model of a front steered vehicle (top view). The four-wheeled vehicle can be modeled as an equivalent two-wheel model. The dynamic equations of the two-wheel car model are well known as follows.

$$mV(\dot{\beta}+\gamma)=2F_f+2F_r \tag{1}$$

$$I_{zz}\ddot{\gamma}=2F_f l_f-2F_r l_r \tag{2}$$

$$F_f=C_f\alpha_f, F_r=C_r\alpha_r \tag{3}$$

$$\alpha_f=\delta_f-(\beta+\frac{l_f}{V}\gamma), \alpha_r=-(\beta-\frac{l_r}{V}\gamma) \tag{4}$$

where,

- m: Vehicle weight
- V: Vehicle velocity
- β: Sideslip angle at center of gravity(COG)
- γ: Yaw rate
- F: Cornering force
- I_{zz}: Vehicle inertia about the vertical axis
- l_f, l_r : Distance from COG to front / rear wheel axis
- C: Cornering power
- α: Sideslip angle of tire
- δ_f: Steering angle

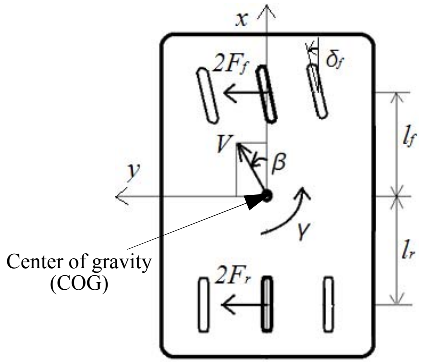


Fig.2 Equivalent two-wheel model

B. *Model of human recognition for lane tracking*

Figure 3 shows a schematic of a car running (top view). A human driver riding on a vehicle recognizes the desired course which the car should approach to. However, the human driver does not look at a position on the desired course just right beside of the vehicle body, $y_r(t)$, but he/she may look a forward position of the vehicle and estimates an error between desired course, $y_r(t+\Delta t)$, and the estimated vehicle position, $y(t+\Delta t)$, at a curtain time Δt later as is shown in Fig.3. The driver might operate a driving device (normally a steering wheel, a joystick lever in this paper) for reducing the recognized error $\varepsilon(t)$. The behavior including a driver recognition and an operating action are represented as follows.

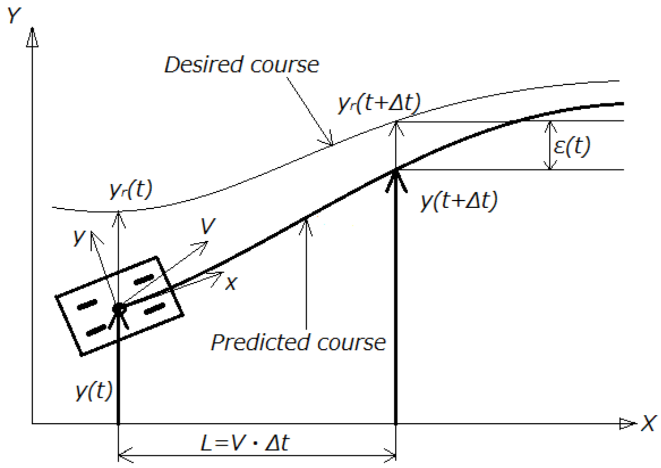


Fig.3 Recognition model of a car driving (2nd order prediction)

$$\delta_j(s) = \frac{K_1}{T_1 \cdot s + 1} \cdot e^{-T_D \cdot s} \cdot \varepsilon(s) \tag{5}$$

$$\begin{aligned} \varepsilon(t) &= y_r(t+\Delta t) - y(t+\Delta t) \\ &= \{y_r(t) - y(t)\} \\ &\quad + \{\dot{y}_r(t) - \dot{y}(t)\}\Delta t \\ &\quad + \{\ddot{y}_r(t) - \ddot{y}(t)\}\frac{\Delta t^2}{2} \end{aligned} \tag{6}$$

Where,

δ_j: Angle of a joystick
K_1: Operation gain
T_1: Time constant
T_d: Time delay
ε: Error
y_r: Y-coordinate of reference course
y: Vehicle position along y-axis

Now we have to consider the influence of a joystick mechanism and a feedback delay which is modeled as a first order transfer function.

$$\delta_{sw}(s) = \frac{K_2}{T_2 \cdot s + 1} \cdot \delta_j(s) \tag{7}$$

where, δ_{sw} is an angle of a steering wheel which is rotated by an electric motor controlled by joystick controller. K_2 and T_2 are the gain and a time delay on the joystick system respectively. The gain K_2 is basically determined by the mechanical conditions of a car and the joystick. The maximum inclination angle of a joystick lever is about 20degs in one side. However, a steering wheel of a car can be operated about 1.5 turns (540degs) in one side. Therefore to rotate the steering wheel to a mechanical limit, K_2 has to be 540/20=27.

However, this value is not appropriate for a car driving even at low speeds, as known in experiences. To reduce the gain around the center region, we introduce a polygonal line as a sensitivity function inserted between a joystick command to steering wheel angle. Figure 4 shows a sensitivity function between a joystick inclination angle to steering wheel angle. In Fig.4, a dotted line shows a linear function which inclination is constant in all regions (K_2=27), while a solid polygonal line which inclination is gradual (K_2=16) around the center but steep at both ends (K_2=49). Since the sensitivity function is calculated by a microcomputer based motor controller, the simple first-order equation such as polygonal line (which can be represented as $y=ax+b$) contributes to maintain system stability and the high frequency feedback calculations.

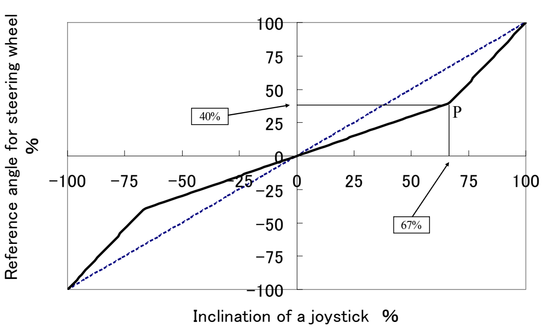


Fig.4 Sensitivity function (joystick to steering wheel)

IV. SIMULATIONS

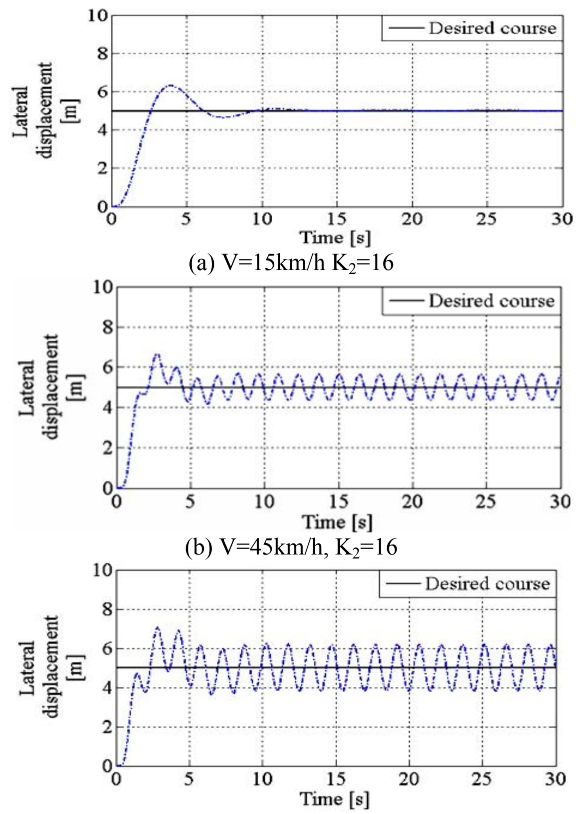
A. Drive by a Joystick system with fixed gain

The behaviors of a car driving by the joystick system are analyzed by a computer simulation using MATLAB/Simulink. The car performances are calculated by eqs(1)-(7) with a parameter set shown in Table1. Note here that the gain of the joystick system is fixed at 16 at all times.

Table 1: Car parameters for simulation

y_r	5 [m]	m	1500[kg]
Y	0 [m]	l_f	1.2 [m]
K_1	0.2 [s]	l_r	1.4 [m]
T_1	0.2 [s]	C_f	40 [kN/rad]
K_2	16 [-]	C_r	60 [kN/rad]
T_2	0.2 [s]	I_{zz}	2.5 [kgm^2]
V	0-60[km/h]	T_d	0.2 [s]

At an initial condition, a car is departed 5m from the reference trajectory line with its orientation is along the X-axis in Fig.3. A car starts to move in respective constant speed to track the desired linear course. Figure 5 shows the performances of lane tracking of a car driven by the joystick system. Figure5(a) shows a performance at 15km/h in which overshoot is about 1m from the reference and converge in 15seconds. Figure5(b) shows a performance at 45km/h which does not converge to the reference and shows a meandering motion continuously. Figure5(c) shows a performance at 60km/h. The amplitude of meandering motion is increased and does not converge to the reference as well.



(a) V=15km/h K_2=16

(b) V=45km/h, K_2=16

(b) V=60km/h K_2=16

Fig.5 Lane tracking simulations with constant joystick gain

B. Drive by a Joystick system with variable gain

It is supposed that the unstable behaviors are considered due to the high gain and the time delay of the joystick system. When a joystick system is installed on a car, we must accept the time delay. Reducing the gain in all ranges of car speed results in unconfort in driving at slow speeds such as less than 10km/h.

To overcome this problem, we analyze an effect of varying the gain based on the car speed. In simulation, we derive the stable gain at respective speed in which an overshoot of the performance is suppressed in 1m from the reference line and converge in 15sec, which is almost the same performance as the one at 15km/h. We determined a siutable gain for a specific vehicle speed by the trial-and-error method on the simulation. Figure 6 shows a result of derived gains. It has to be reduced dramatically between 15km/h to 30km/h, while gradually decreased after 30km/h as shown in Fig.6.

Figure 7 shows performances of a car with a variable joystick gain, based on the plot in Fig.6. It is seen that the stable drive performances are maintained till 60km/h.

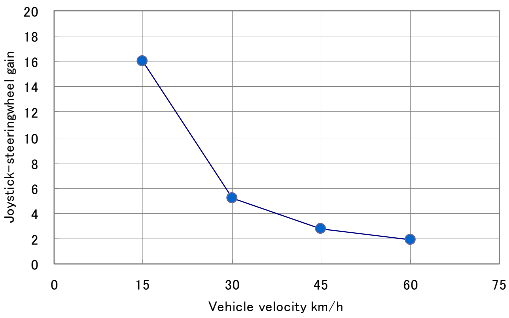
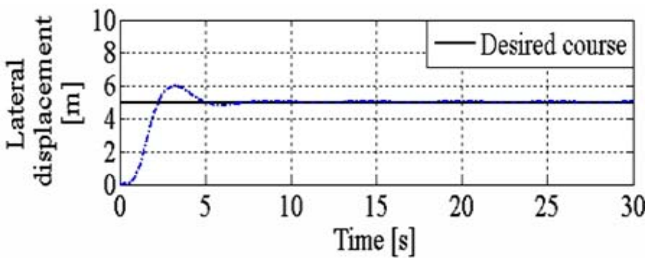
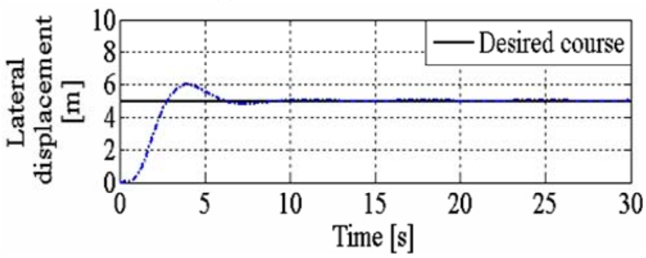


Fig.6 Variable gain for stable car driving



(a) V=45km/h, K2=2.8



(b) V=60km/h, K2=1.9

Fig.7 Lane tracking simulations with variable joystick gains

V. CONCLUSIONS

In this paper, we have analyzed driving stability of a car driven by a joystick interface. For the analysis, we built the model including a car dynamics, human driver recognitions and drive actions and a dynamics of the joystick system. By the simulations, it is clarified that a car driven using a joystick system becomes unstable in high speed drive due to the high gain and the time delay of the joystick system.

To overcome the problem, we proposed a variable gain method in which the joystick gain is varied based on the vehicle speed. The stable gains at each speed were successfully derived and stable performances were realized in high speed drive such as at 60km/h.

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